

## **Department of Information Science and Engineering**

### **Democratizing Medical Diagnosis: A Visual Question Answering (VQA) based Approach**

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**Existing Diagnostic Decision Support Systems (DDSS)** either take:

- a **clinical image** (CT scans, X-Rays, etc.) as an input or,
- a **natural-language question** as an input

so as **to perform a specific task** like, cancer classification (via images) or a simple symptom checker chatbot (via natural language).

Although such systems are effective, **clinicians often need to ask questions specific to a clinical image.**

- *“what is the organ shown in this x-ray scan?”*
- *“what abnormality is seen in the image?”*
- *“in what plane is this image?”*

For such use-cases, there is a **need for a system which takes into account both - the image as well as the natural-language questions** of the clinicians.

# Problem Statement

- The proposed work is to build a **Diagnostic Decision Support System (DDSS)** which **takes as input a clinical image and a natural-language question about the image**, and produces a natural language answer as the output.
- **DDSS employs a Visual Question Answering (VQA) system** which works at intersection of computer vision and natural-language processing techniques.
- A system of this caliber would be able to **assist clinicians and the general public** in answering their complicated queries.
- With this system, the project aims to take a step forward in **democratizing medical diagnosis** and providing equal healthcare opportunities for all.

# VQA as a Confirmatory Step in Diagnosis



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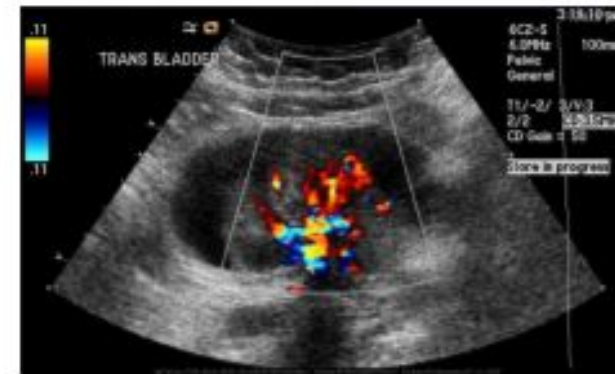
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We can define a VQA System as

“An algorithm that takes as input an image and a natural language question about the image and generates a natural language answer as the output”



(d) **Q:** what plane is this?  
**A:** lateral



(a) **Q:** what imaging method was used? **A:** us-d - doppler ultrasound

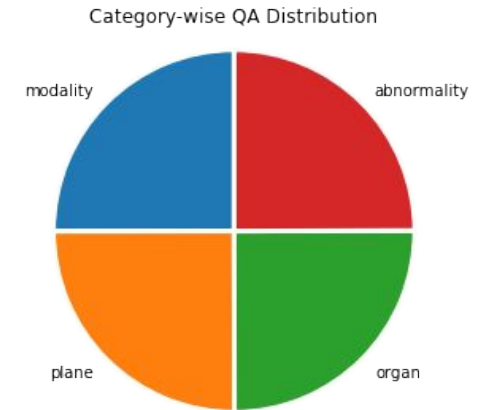
# Exploring ImageCLEF 2019 Dataset



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|       | questions                                 | answers                                           |
|-------|-------------------------------------------|---------------------------------------------------|
| 1287  | what modality is used to take this image? | xr - plain film                                   |
| 4487  | what plane is demonstrated?               | lateral                                           |
| 7687  | what is the organ system in this image?   | breast                                            |
| 10887 | what is abnormal in the x-ray?            | topic: lumbar instability: functional and anat... |



**Each image has a total of 4 questions associated with it.**

**All 4 questions belong to different categories i.e. the 4 categories identified - Modality, Plane, Organ and Abnormality categories.**

# Exploring ImageCLEF 2019 Dataset



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Image Repository

```
synpic41148|what kind of image is this?|cta - ct angiography
synpic43984|is this a t1 weighted image?|no
synpic38930|what type of imaging modality is used to acquire the image?|us - ultrasound
synpic52143|is this a noncontrast mri?|no
synpic20934|what type of image modality is this?|xr - plain film
synpic19141|with what modality is this image taken?|xr - plain film
synpic25502|what was this image taken with?|xr - plain film
synpic38317|is this a contrast or noncontrast ct?|noncontrast
synpic22829|was the ct scan taken with contrast?|yes
synpic39477|what is the mr weighting in this image?|t2
synpic54866|what type of imaging modality is shown?|ct w/contrast (iv)
synpic60053|what type of imaging does this represent?|xr - plain film
synpic28004|is this a ct scan?|yes
synpic51949|is this an mri image?|no
synpic45041|was gi contrast given to the patient?|no
synpic13026|was gi contrast given to the patient?|no
synpic52613|what was this image taken with?|xr - plain film
synpic21981|is this a ct scan?|no
synpic53380|what imaging modality was used to take this image?|mr - flair
synpic21144|what modality was used to take this image?|mammograph
synpic20556|is this an mri image?|no
synpic39156|is this a contrast or noncontrast mri?|noncontrast
synpic41586|is this a ct scan?|yes
synpic36892|what imaging modality was used to take this image?|ct with iv contrast
synpic20868|is this a ct scan?|yes
synpic27646|what type of image modality is this?|xr - plain film
synpic21181|what type of image modality is seen?|mammograph
synpic19981|was gi contrast given to the patient?|no
synpic54237|what imaging modality was used to take this image?|mr - flair
synpic33704|what type of imaging is this?|us - ultrasound
synpic43008|is this a ct scan?|no
synpic23493|is this a t1 weighted image?|yes
```

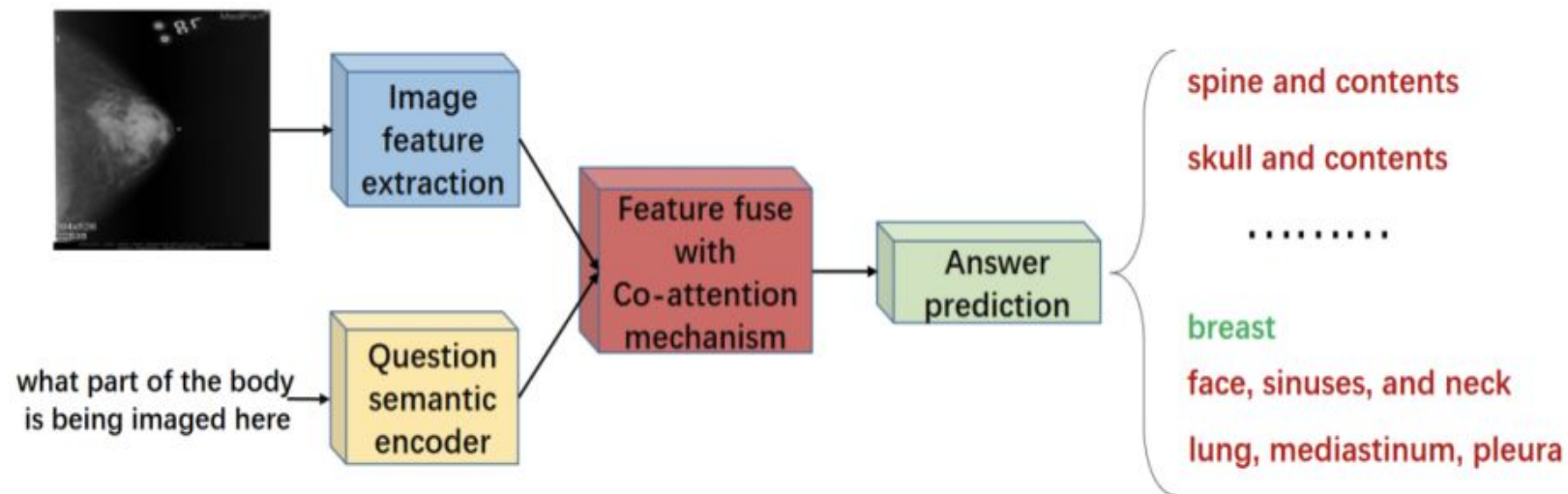
QA Pairs

# Approaches Explored

- Throughout the entirety of this project, **2 approaches were explored** and their findings were noted.
- The 2 approaches are as follows:
  - **Approach 1:** Hierarchical Question-Image Co-attention model (***Primary Focus***)
  - **Approach 2:** Multi-modal Factorized Bilinear Pooling model with Co-attention

The proposed solutions consists of 3 modules

- Image Feature Extraction
- Question Semantic Encoder
- Feature Fusion with Co-attention mechanism





# **Approach 1:**

## **Hierarchical Question-Image Co-attention model**

# Objectives for Approach 1

- Employ a ResNet18 network to extract the inherent features from image data.
- Extracting semantic features from the question using LSTM.
- Fusing the features from previous two modules to predict answers for VQA model.
- Training and Fine tuning VQA model to obtain enhanced accuracy.

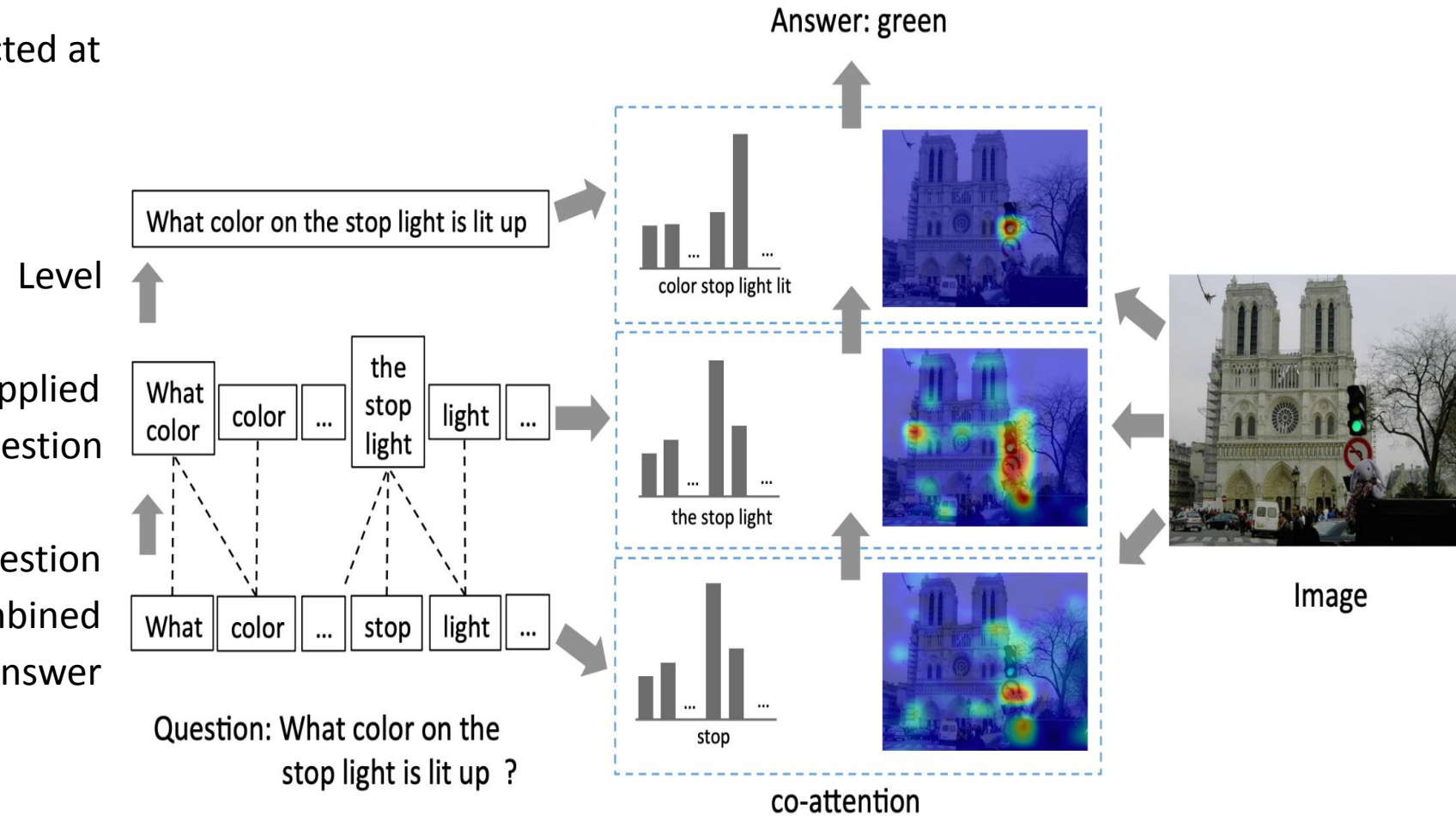
# Hierarchical Question Representation



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- Question embeddings are extracted at 3 levels:
  - Word Level
  - Phrase Level
  - Sentence
- At each level, co-attention is applied on both the image and question
- These co-attended image-question attention maps are then combined together to get the final answer prediction.



- Parallel co-attention **attends to the image and question simultaneously.**
- Parallel co-attention is done at each level in the hierarchy.
- **Image and question are connected by calculating the similarity** between image (**V**) and question (**Q**) features at all pairs of image-locations and question-locations.
- This **affinity matrix** (**C**) is calculated as:

$$C = \tanh(Q^T W_b V)$$

- Conventionally, **activation is calculated by computing the max()** of all the affinity matrices:

$$a^v[\bar{n}] = \max_i(C_{i,n}) \text{ and } a^q[\bar{t}] = \max_j(C_{t,j})$$

- But here, we consider the affinity matrix as a feature, and then predict image and question attention feature maps as:

$$\mathbf{H}^v = \tanh(\mathbf{W}_v \mathbf{V} + (\mathbf{W}_q \mathbf{Q}) \mathbf{C}), \quad \mathbf{H}^q = \tanh(\mathbf{W}_q \mathbf{Q} + (\mathbf{W}_v \mathbf{V}) \mathbf{C}^T)$$

$$\mathbf{a}^v = \text{softmax}(\mathbf{w}_{hv}^T \mathbf{H}^v), \quad \mathbf{a}^q = \text{softmax}(\mathbf{w}_{hq}^T \mathbf{H}^q)$$

- $\mathbf{a}^v$  and  $\mathbf{a}^q$  are the attention probabilities for each image region  $\mathbf{v}^n$  and word  $\mathbf{q}^t$  respectively.
- The affinity matrix  $\mathbf{C}$  transforms question attention space to image attention space (vice versa for  $\mathbf{C}^T$ ).
- Finally, the image and question attention vectors are calculated as the weighted sum of the image features and question features:

$$\hat{\mathbf{v}} = \sum_{n=1}^N a_n^v \mathbf{v}_n, \quad \hat{\mathbf{q}} = \sum_{t=1}^T a_t^q \mathbf{q}_t$$

- VQA is treated as a **classification task**.
- The answer prediction is based on **co-attended image and question from all 3 levels**.
- A **multi-layer perceptron (MLP)** is used to recursively encode the attention features and get '**p**' probability of the final answer.

$$\mathbf{h}^w = \tanh(\mathbf{W}_w(\hat{\mathbf{q}}^w + \hat{\mathbf{v}}^w))$$

$$\mathbf{h}^p = \tanh(\mathbf{W}_p[(\hat{\mathbf{q}}^p + \hat{\mathbf{v}}^p), \mathbf{h}^w])$$

$$\mathbf{h}^s = \tanh(\mathbf{W}_s[(\hat{\mathbf{q}}^s + \hat{\mathbf{v}}^s), \mathbf{h}^p])$$

$$\mathbf{p} = \text{softmax}(\mathbf{W}_h \mathbf{h}^s)$$

- The ImageCLEF 2019 dataset has been divided into ***Train-Test splits***.
- Dataset is loaded into the model using ***“Data Loaders”*** for both - testing and training.
- **Training dataset size:** 3,700 images and 14,800 QA-pairs
- **Testing dataset size:** 501 images and 2,004 QA-pairs
- For each epoch during training, the model is trained over the training dataset and tested over the test dataset
  - An accuracy score as well as cross-entropy loss is computed for each epoch.
- Formula for calculating accuracy:

$$\text{accuracy} = \frac{\text{Total number of successful classification}}{\text{Length of data loader} * \text{Batch size}}$$



# Testing



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- The initial model, trained for 200 epochs, gave **erroneous accuracy scores and loss values**



```
stats.txt
{"epoch": 175, "loss": 3.088129997253418, "time": 540}
{"epoch": 176, "loss": 2.5887038707733154, "time": 541}
{"epoch": 177, "loss": 3.444145917892456, "time": 541}
{"epoch": 178, "loss": 1.625876545906067, "time": 541}
/content/drive/MyDrive/MedVQA/hierarchical/main.py
{"epoch": 179, "loss": 4.631123065948486, "time": 2283}
{"epoch": 180, "loss": 3.2780849933624268, "time": 540}
{"epoch": 181, "loss": 4.05740213394165, "time": 540}
{"epoch": 182, "loss": 3.554409980773926, "time": 541}
{"epoch": 183, "loss": 3.328317403793335, "time": 540}
{"epoch": 184, "loss": 3.791888475418091, "time": 540}
{"epoch": 185, "loss": 4.455935478210449, "time": 540}
/content/drive/MyDrive/MedVQA/hierarchical/main.py
{"epoch": 186, "loss": 5.468970775604248, "time": 1622}
/content/drive/MyDrive/MedVQA/hierarchical/main.py
{"epoch": 187, "loss": 5.153775691986084, "time": 1150}
{"epoch": 188, "loss": 4.843677997589111, "time": 316}
{"epoch": 189, "loss": 4.692816257476807, "time": 318}
{"epoch": 190, "loss": 4.439323902130127, "time": 317}
{"epoch": 191, "loss": 3.2392642498016357, "time": 317}
{"epoch": 192, "loss": 4.06124210357666, "time": 318}
{"epoch": 193, "loss": 2.8026487827301025, "time": 317}
{"epoch": 194, "loss": 2.4563934803009033, "time": 317}
{"epoch": 195, "loss": 3.8541557788848877, "time": 317}
{"epoch": 196, "loss": 4.282835006713867, "time": 317}
{"epoch": 197, "loss": 4.452075481414795, "time": 317}
{"epoch": 198, "loss": 3.661985397338867, "time": 318}
{"epoch": 199, "loss": 2.5803303718566895, "time": 317}
/content/drive/MyDrive/MedVQA/hierarchical/main.py
```



# Testing

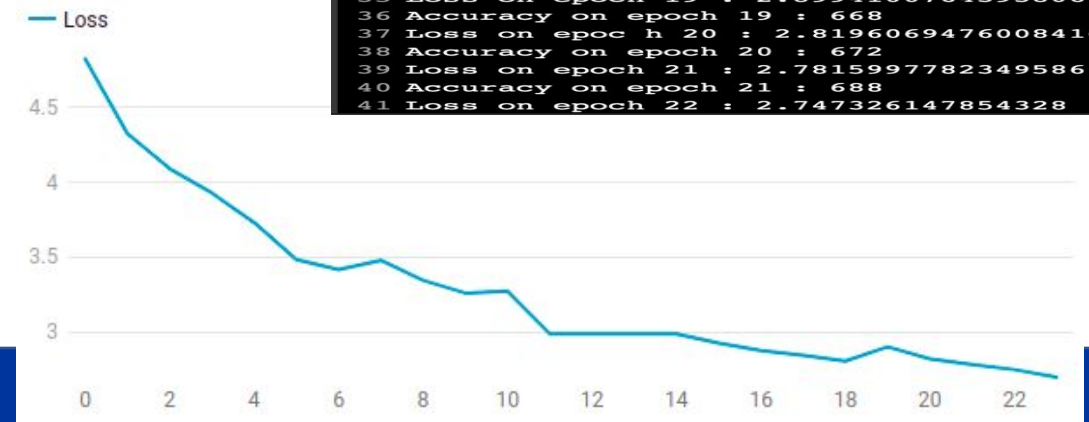


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- A new model was then put to training with the following changes:
  - Added a **scheduler to implement learning rate decay**
  - Set the **patience to 2** (default: 3)
- The results led to an **accuracy score of 42%** and a **cross-entropy loss of 1.67301**
- The intermediate results of this model show a **steady descent in the loss curve** with a much **improved accuracy score**.

```
model.py X loss_of_epoch1.txt X
/content/model.py
1 Loss on epoch 0 : 4.827191402316093
2 Accuracy on epoch 0 : 363
3 Loss on epoch 1 : 4.327024203538895
4 Accuracy on epoch 1 : 428
5 Loss on epoch 2 : 4.091721806526184
6 Accuracy on epoch 2 : 473
7 Loss on epoch 3 : 3.9289184272289277
8 Accuracy on epoch 3 : 483
9 Loss on epoch 4 : 3.7313357794284823
10 Accuracy on epoch 4 : 548
11 Loss on epoch 5 : 3.483756142258644
12 Accuracy on epoch 5 : 591
13 Loss on epoch 6 : 3.417079070806503
14 Accuracy on epoch 6 : 550
15 Loss on epoch 7 : 3.4783317309618
16 Accuracy on epoch 7 : 566
17 Loss on epoch 8 : 3.3442553249001503
18 Accuracy on epoch 8 : 641
19 Loss on epoch 9 : 3.259169156849384
20 Accuracy on epoch 9 : 606
21 Loss on epoch 10 : 3.2722210550308226
22 Accuracy on epoch 10 : 548
23 Loss on epoch 11 : 3.136818144917488
24 Accuracy on epoch 11 : 652
25 Loss on epoch 14 : 2.986817311346531
26 Accuracy on epoch 14 : 645
27 Loss on epoch 15 : 2.924548974633217
28 Accuracy on epoch 15 : 639
29 Loss on epoch 16 : 2.8756509897112847
30 Accuracy on epoch 16 : 665
31 Loss on epoch 17 : 2.8414809301495554
32 Accuracy on epoch 17 : 672
33 Loss on epoch 18 : 2.8038976994156837
34 Accuracy on epoch 18 : 700
35 Loss on epoch 19 : 2.8994100764393806
36 Accuracy on epoch 19 : 668
37 Loss on epoch 20 : 2.8196069476008416
38 Accuracy on epoch 20 : 672
39 Loss on epoch 21 : 2.7815997782349586
40 Accuracy on epoch 21 : 688
41 Loss on epoch 22 : 2.747326147854328
```



# Results for Approach 1

Our contributions to this domain can be summarised in 4 points:

- Shifted the architecture's focus from general VQA to MedVQA domain (i.e. MedVQA radiology dataset)
- Implemented an answer dictionary for training the model over MedVQA dataset
- Implemented a scheduler function or learning rate decay technique

# **Approach 2:**

## **Multi-modal Factorized Bilinear Pooling model with Co-attention**

# Objectives for Approach 2

- Building a VGG16 network to extract the inherent features from image data.
- Extracting semantic features from the question using BERT.
- Fusing the features from previous two modules using MFB to predict answers for VQA model.

# Image Feature Extraction

**Image Feature Extraction:** The proposed model uses a new convolution network that is based on the **VGG16 network (pre-trained on ImageNet)** and **Global Average Pooling Strategy**

- Input to the model is an image of size  $224 \times 224 \times 3$ .
  - Use the VGG16 model pretrained on the ImageNet dataset.
  - Extract features of all 6 blocks from VGG16.
  - Perform Global Average Pooling (GAP) on the extracted Feature tensors individually.
  - Concatenate all the resulting tensors after performing GA
- Resultant tensor obtained is of  $1 \times 1 \times 1984$  dimensions.
- **Outcome:** the new network structure could effectively avoid overfitting and improve the accuracy of the model.

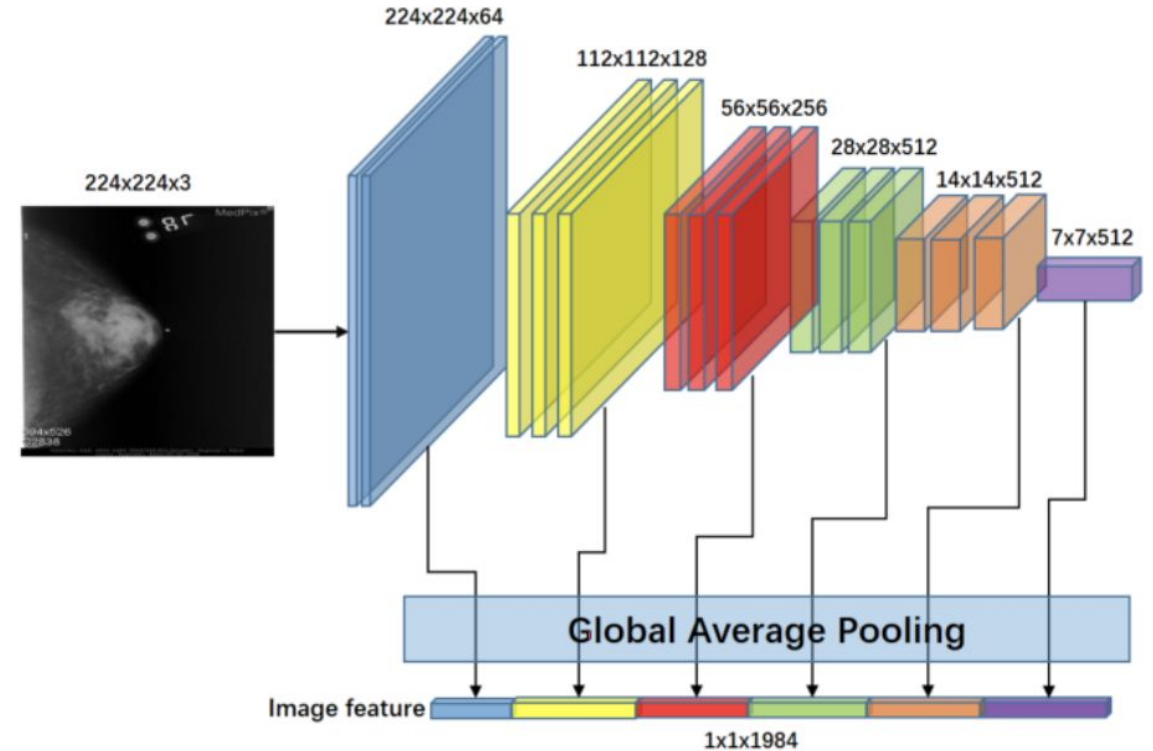


Fig. 3. Image feature extraction

# Question Semantic Encoder



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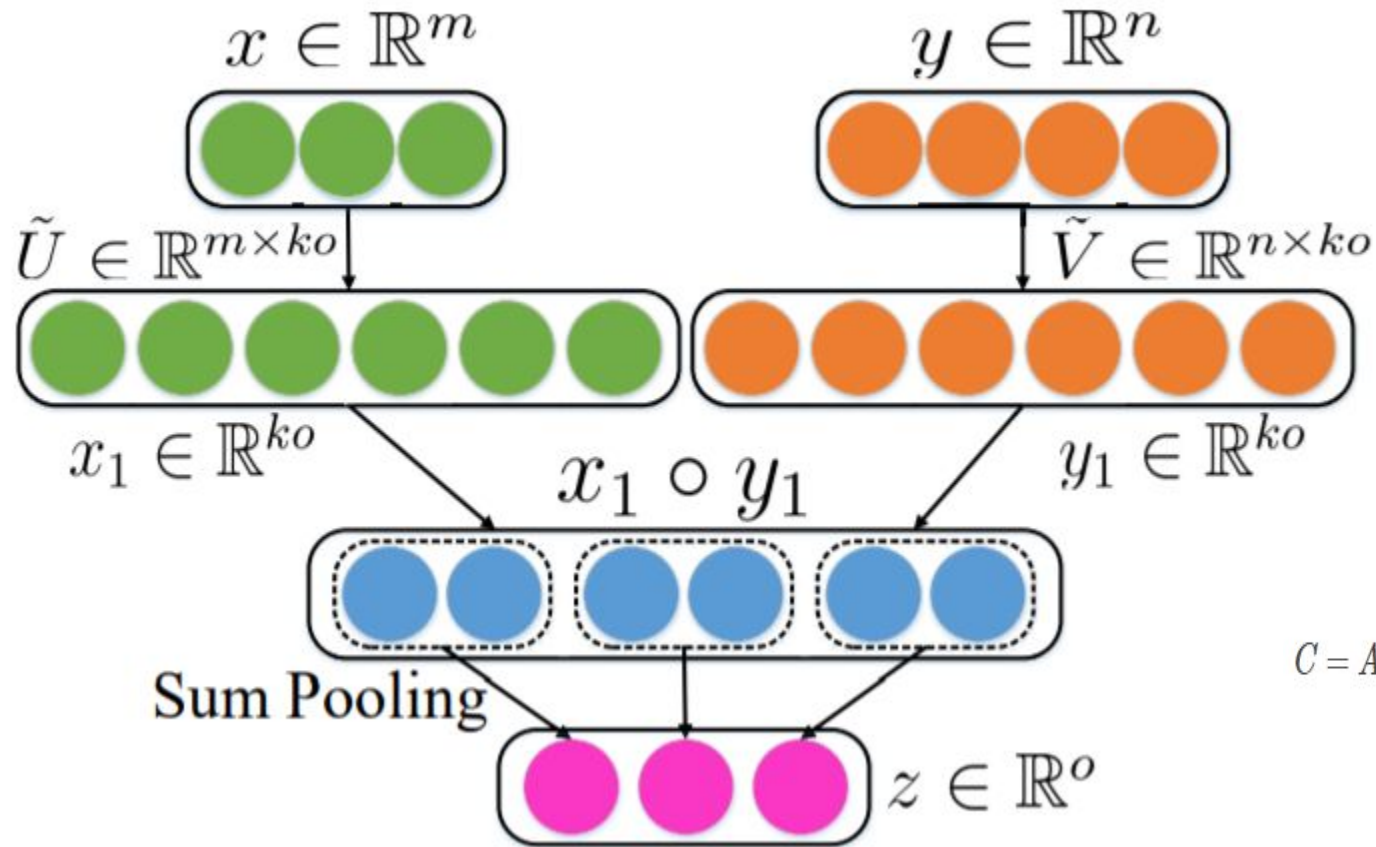
- The “**base BERT**” model is used here - which includes 12 layers and 768 hidden variables with a total of 110M parameters - to get the semantic feature of the question.
- Bidirectional Encoder Representations from Transformers (BERT), is a Transformer-based machine learning technique for natural language processing pre-training development by Google.
- Bert uses Transformers - an attention mechanism that allows the model to learn contextual relations between words in a text.
- BERT is a “*deeply bidirectional*” model. Bidirectional means that BERT learns information from both the left and the right side of a token’s context during the training phase.
- The features are extracted from the last and penultimate layers , average of both these features obtained result in a 768 dimensional tensor

# MFB Internal Architecture



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$$C = A \circ B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \circ \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} & a_{13}b_{13} \\ a_{21}b_{21} & a_{22}b_{22} & a_{23}b_{23} \\ a_{31}b_{31} & a_{32}b_{32} & a_{33}b_{33} \end{bmatrix}$$

**Module 1:** Successfully built a VGG-16 network to extract the inherent features from image data and obtained a 1x1x1984 dimensional vector

```
[ ] result = np.concatenate((block1_gap, block2_gap, block3_gap, block4_gap, block5_gap, block6_gap), axis=1)
result = np.array([result])
print(result.shape)
print(result)
```

```
(1, 1, 1984)
[[[141.51372  123.1967  121.80125  ...  8.5662775  0.3047954
    12.977199 ]]]
```



**Module 2:** Successfully Extracted semantic features from the question using "base BERT" and obtained a 768 dimensional tensor .

```
▶ print(result1.shape)
  print(result1)

↳ (1, 128, 768)
  [[[-0.78652704  0.32944506 -0.63347924 ... -0.5159604 -0.02301432
      0.26579255]
    [ 0.9281898  1.3219947  0.17934746 ... -0.42555523  0.44262075
     -0.17138335]
    [-0.72876966  0.58277035  0.01902795 ...  0.18465838 -0.00838665
     -0.49491024]
    ...
    [ 0.47007158  0.6210689  0.5784095  ... -0.37159032 -0.18816018
     0.30970842]
    [-0.05151443  0.60759974  0.53674793 ... -0.30779076 -0.4174086
     0.37087354]
    [-0.20718032  0.25423297  0.3103047  ... -0.51423943 -0.38613734
     0.2928851  ]]]
```

**Module 3:** Successfully combined the outputs from the previous two modules using Multi-Modal Bilinear Fusion.

```
▶ print(mfb_output.shape)  
print(mfb_output)
```

```
↳ torch.Size([5, 1000])  
tensor([[0.0010, 0.0010, 0.0011, ..., 0.0010, 0.0010, 0.0010],  
        [0.0009, 0.0010, 0.0010, ..., 0.0010, 0.0010, 0.0010],  
        [0.0010, 0.0010, 0.0010, ..., 0.0010, 0.0009, 0.0010],  
        [0.0010, 0.0011, 0.0011, ..., 0.0010, 0.0010, 0.0010],  
        [0.0010, 0.0009, 0.0010, ..., 0.0010, 0.0010, 0.0010]],  
        grad_fn=<SoftmaxBackward0>)
```

- In **Approach 1:**

- LSTM is advantageous because it resolves the vanishing gradient problem.
- Hierarchical Image-Question Co-attention model provides a relatively simpler architecture than MFB to fuse the extracted features.
- The Hierarchical Question representation enables a deeper understanding of the question-text with Co-Attention applied at all 3 levels of the hierarchy.

- In **Approach 2:**

- Global Average Pooling is employed in the custom VGG16 architecture to resolve the problem of overfitting.
- MFB is amongst the state-of-the-art techniques for merging multiple features together.

The logo of M. V. V. Meenakshi Institute of Technology is a circular emblem. It features a central globe with a torch and an open book. The text "M. V. V. MEENAKSHI INSTITUTE OF TECHNOLOGY" is written around the top half of the circle, and "VALLUUR" is at the bottom. A laurel wreath is at the base of the emblem.

# Thank you

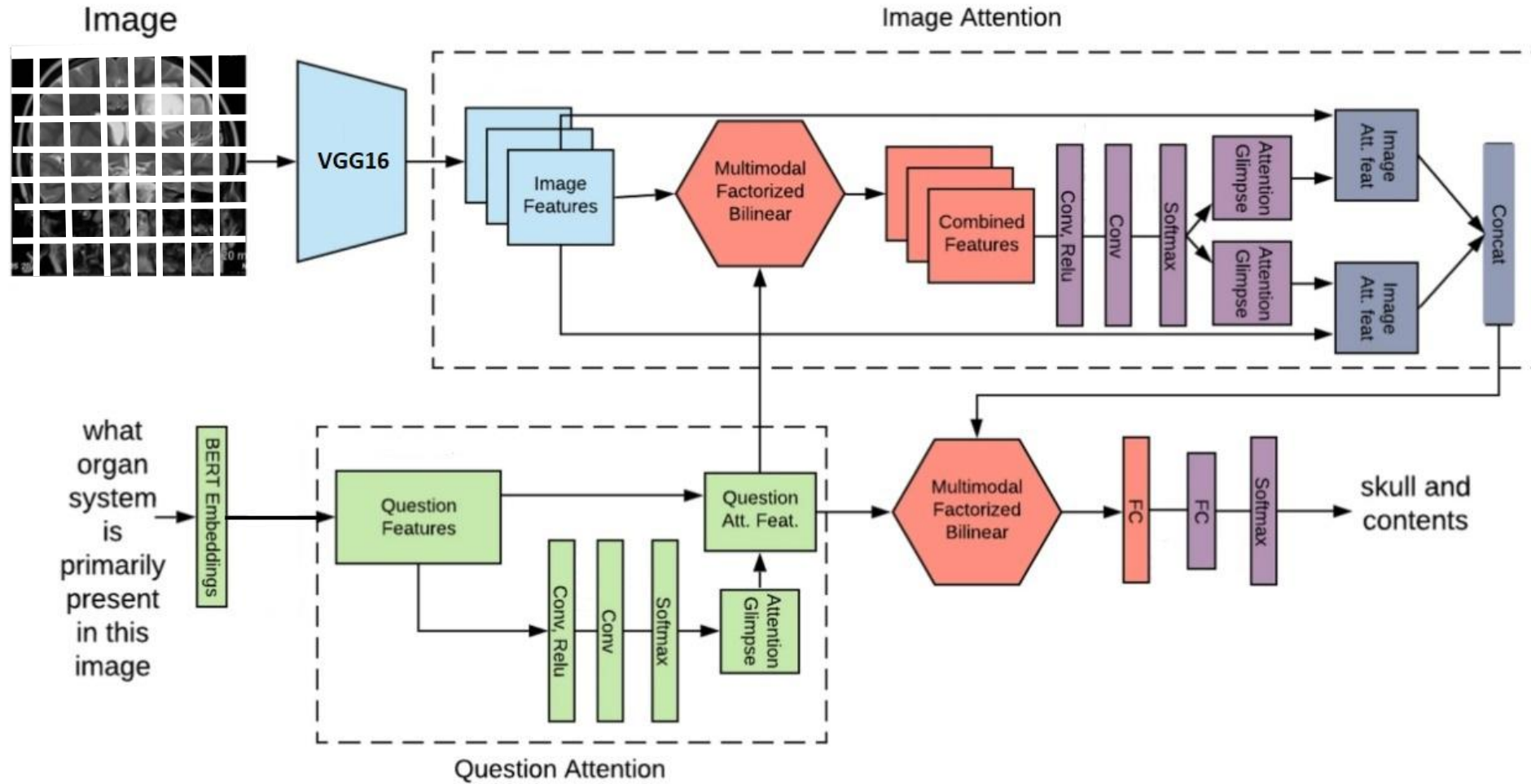
KNOWLEDGE • CHARACTER • UNITY

# Feature Fusion with Co-attention mechanism



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- **Training the model and fine tuning** to obtain the final result
- **Experimenting with different algorithms** to obtain better accuracy

- **Medical images are difficult to comprehend** for a person without expertise.
- The **scarcity of medical practitioners across the globe causes fatigue due to the high number of cases**, inducing human errors during the diagnosis.
- In such scenarios, **having a 'second opinion' can be helpful** in boosting the confidence of the decision maker.

This gap can be bridged using Diagnostic Decision Support Systems (DDSS) but current DDSS are limited in their use-cases since they cannot answer complex, image-oriented questions of the clinician.



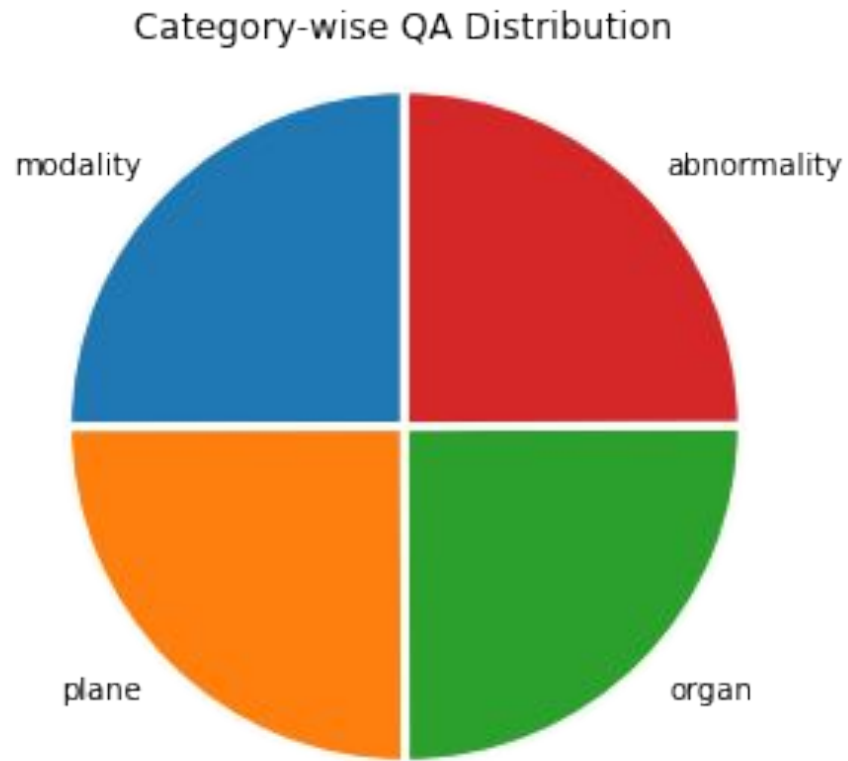


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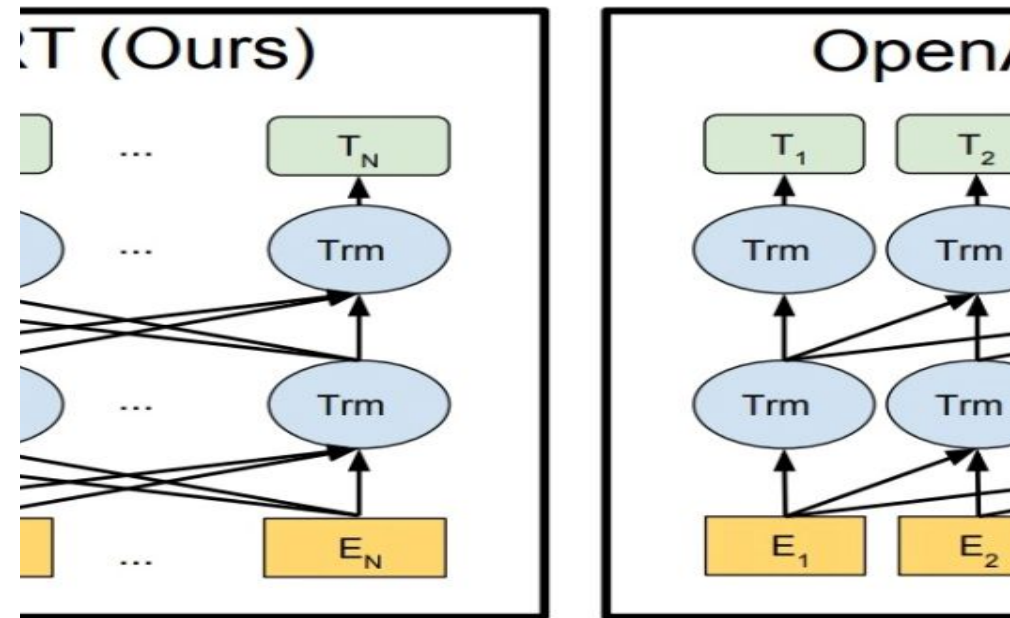
steps for implementing the **Question Semantic Encoder** module:-

- Use the "BERT Base" model.

- Extract features from the last and penultimate layers.

- Average both the features which should result in a 768-dimensional tensor.

**Feature fuse with co-attention mechanism:** It is the strategy to combine visual and semantic features. “We use multi-modal factorized bilinear pooling (MFB) with co-attention to fuse the two modalities of features.



# Objectives

- To **replicate the results of the current state-of-the-art VQA model** for the ImageCLEF 2019 dataset. ([Yan, Xin, et al. "Zhejiang University at ImageCLEF 2019 Visual Question Answering in the Medical Domain." CLEF 2019](#))
- Make improvements over the current state-of-the-art model by:
  - Experimenting between various natural-language processing models (**BioBERT or SciBERT**)
  - Experimenting with VGG-19 for image feature extraction ([Chen, Guangqi, Haifan Gong, and Guanbin Li. "Hcp-mic at vqa-med 2020: Effective visual representation for medical visual question answering." CLEF, 2020.](#))
  - Experimenting with different data augmentation techniques (**[ZCA whitening](#), image rotation, zoom, etc.**)

# Problem Statement

Strong reasons as to why vqa is needed , therefore proposing a vqa , how will it help the society for good

Current Diagnostic Decision Support Systems either take an image or a natural-language question as an input so as to perform a specific task like, cancer classification (via images) or a symptom-checker(//or verification) chatbot (via natural language). Though such systems are useful, clinicians often need to ask questions specific to an image. For such use-cases, there is a need for a system which takes into account both - the image as well as the natural-language questions of the clinicians.

In the domain of healthcare, clinicians often use radiology scans of their patients for diagnosis purposes. They, therefore, require intelligent DDSS which can answer questions specific to a clinical image (like, CT scans, X-Rays, etc.). For such use-cases, there is a need for a system which takes into account both - the image as well as the natural-language questions of the clinicians and works at intersection of computer vision and natural-language processing techniques.